

# Zubair Ahmed

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## Technical Skills

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**Skills:** Data Science, Machine Learning, Deep Learning, Data Analysis.

**Libraries/Tools:** Pandas, NumPy, Rasterio, Geopandas, GDAL, Xarray, Dask, Matplotlib, Seaborn, Keras, Tensorflow, PyTorch, huggingface, scikit-learn, LangGraph, LangChain, fastapi, QGIS, ArcGIS, Google Earth Engine, Revit, CAD.

**Languages:** Python, Javascript (GEE).

**Developer Tools:** Git, Visual Studio Code, Jupyter Notebook, Docker, DVC.

## Education

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**Karnataka State Remote Sensing Applications Centre - VTU**

Bangaluru, KA

M.tech in Geoinformatics; Remote sensing and GIS

Jan. 2020 – Jul. 2022

**9.23 CGPA**

**HKBK college of Engineering - VTU**

Bangaluru, KA

B.tech in Civil Engineering

Jan. 2015 – May. 2019

**8.34 CGPA**

## Experience

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### **Data Scientist**

Bangalore, KA

Captain Fresh.

Aug. 2022 – Present

#### **Geospatial Data Engineering**

- Created scalable Data Extraction pipeline using Xarray and Dask for extracting MSI level-2A dataset from AWS open data registry and Planetary computer

#### **Semantic Segmentation, MLOps & Deployment**

- Implemented semantic segmentation models (U-Net, DeepLabv3plus, U<sup>2</sup>-Net, and SAM) with ResNet and ViT backbones to delineate aquaculture pond boundaries, achieving 94% mIoU across four geographically distinct regions.
- Developed a DVC pipeline for versioning and tracking of code, data, and models.
- Docker based containerisation of code and models for deployment.

#### **Object detection**

- Finetuned YOLOv8 on high-resolution SAR imagery for micro-target detection.

#### **Digital Image processing and Machine Learning**

- Correlated satellite-derived surface reflectance with in-situ water quality parameters using regression techniques, achieving an average R<sup>2</sup> of 0.80.
- Developed a species specific index called as Shrimp Index for highlighting the characteristics of shrimps and a Pond Activity Indicator (PAI) for classifying the status of the pond through scientific visualisation of spectral signatures.
- Developed rule based RS driven Early Warning System for shrimp ponds.

### **Geoinformatics Intern**

Bangalore, KA

KSRASAC.

Sep. 2021 – Nov. 2021

#### **Learnings from internship are:**

- Visual Image interpretation and delineation of features from the satellite imageries based on NRSC LULC schema.
- Filtering of single image from image collection based on metadata using google earth engine.
- Interpretation of multispectral, thermal and SAR data from google earth engine.
- Ground truthing and performing supervised classification on various band combinations.
- Geo spatial analysis using software's like. QGIS, ArcGIS and ERDAS Imagine.

### **Site Engineer (Revit - Architecture, CAD)**

Bangalore, KA

April. 2019 – Mar 2020

- Developed floor plan drawings based on project requirements and site constraints.
- Prepared detailed structural drawings in coordination with design specifications.
- Created architectural elevations ensuring design accuracy and compliance
- Executed on-site layout activities, including centerline marking and alignment verification
- Assisted in translating design drawings into practical site execution workflows

## Projects

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- Production Estimation of Horticulture Crops using U-net** | Python, google Earth Engine, Tensorflow, Sklearn, Rasterio, Geopandas. Bangalore, KA  
Apr. 2022 – Aug. 2022
- Applied U-net and Random Forest models on Sentinel-2 imagery (GEE & Colab) for acreage and production estimation of mango and coconut crops.
  - Achieved 84% accuracy with Unet using NIR, Red, and Green bands, outperforming RF and conventional approaches.
  - Demonstrated transfer learning success by applying the trained CNN model across different study areas with encouraging results.

## Projects

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**LULC Classification using Deep Neural Networks** | Python, google Earth Engine, Tensorflow, Sklearn, Rasterio, Geopandas. Bangalore, KA  
Dec. 2021 – Mar. 2022

Developed and trained Random Forest (RF) and Deep Neural Network (DNN) models for accurate classification of diverse land cover features in Mandya District.

**Soil Stabilization using Class 'C' Fly Ash and Bacteria** Bangalore, KA  
2019

In this study, fly ash combined with soil bacteria was evaluated over different curing periods to determine its effect on the soil's bearing capacity

## Research Papers

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**Area Estimation of Mango and Coconut Crops using Machine Learning in Hesaraghatta Hobli of Bengaluru Urban District, Karnataka [Paper]** Bangalore, KA  
April. 2023

In this research, U-Net with a ResNet34 backbone and Random Forest algorithms were applied to Sentinel-2 imagery for estimating mango and coconut crop areas in Hesaraghatta, Bengaluru. The U-Net model using NIR, Red, and Green bands achieved 84% accuracy, outperforming RF and conventional methods, and demonstrated promising transfer learning potential across different study areas.